CRAMER'S RULE

LINEAR ALGEBRA

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Cramer's Rule

CRAMER'S RULE

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_3 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$A_x = \begin{bmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_3 \\ d_3 & b_3 & c_3 \end{bmatrix}$$

$$A_y = \begin{bmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_3 \\ a_3 & d_3 & c_3 \end{bmatrix}$$

$$A_z = \begin{bmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_3 \\ a_3 & b_3 & d_3 \end{bmatrix}$$

$$A_x = \begin{bmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_3 \\ d_3 & b_3 & c_3 \end{bmatrix}$$

$$x = \frac{|A_x|}{|A|} \qquad y = \frac{|A_y|}{|A|} \qquad z = \frac{|A_z|}{|A|}$$

$$A_y = \begin{bmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_3 \\ a_3 & d_3 & c_3 \end{bmatrix}$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_3 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$A_z = \begin{bmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_3 \\ a_3 & b_3 & d_3 \end{bmatrix}$$

EXERCISE

Solve the given system of linear equations using Cramer's Rule.

$$4a + 4b + 7c = 5.2$$

$$6a + 4b + 8c = 6.2$$

$$5a + 5b + 4c = 4.6$$

Solve the given system of linear equations using Cramer's Rule.

$$4a + 4b + 7c = 5.2$$

$$6a + 4b + 8c = 6.2$$

$$5a + 5b + 4c = 4.6$$

$$M = \begin{bmatrix} 4 & 4 & 7 \\ 6 & 4 & 8 \\ 5 & 5 & 4 \end{bmatrix}$$

$$M_a = \begin{bmatrix} 5.2 & 4 & 7 \\ 6.2 & 4 & 8 \\ 4.6 & 5 & 4 \end{bmatrix}$$

$$M_b = \begin{bmatrix} 4 & 5.2 & 7 \\ 6 & 6.2 & 8 \\ 5 & 4.6 & 4 \end{bmatrix}$$

$$M_c = \begin{bmatrix} 4 & 4 & 5.2 \\ 6 & 4 & 6.2 \\ 5 & 5 & 4.6 \end{bmatrix}$$

$$|M| = (4 \cdot 4 \cdot 4) + (4 \cdot 8 \cdot 5) + (7 \cdot 6 \cdot 5)$$

$$-(5 \cdot 4 \cdot 7) - (5 \cdot 8 \cdot 4) - (4 \cdot 6 \cdot 4)$$

$$|M| = 434 - 396$$

$$|M| = 38$$

$$M = \begin{bmatrix} 4 & 4 & 7 \\ 6 & 4 & 8 \\ 5 & 5 & 4 \end{bmatrix} \begin{bmatrix} 4 & 4 & 7 \\ 6 & 4 & 8 \\ 5 & 5 & 4 \end{bmatrix}$$

$$\underline{|M| = 38}$$

$$M_a = \begin{bmatrix} 5.2 & 4 & 7 \\ 6.2 & 4 & 8 \\ 4.6 & 5 & 4 \end{bmatrix}$$

$$M_b = \begin{bmatrix} 4 & 5.2 & 7 \\ 6 & 6.2 & 8 \\ 5 & 4.6 & 4 \end{bmatrix}$$

$$M_c = \begin{bmatrix} 4 & 4 & 5.2 \\ 6 & 4 & 6.2 \\ 5 & 5 & 4.6 \end{bmatrix}$$

$$|M_a| = (5.2 \cdot 4 \cdot 4) + (4 \cdot 8 \cdot 4.6) + (7 \cdot 6.2 \cdot 5)$$

$$-(4.6 \cdot 4 \cdot 7) - (5 \cdot 8 \cdot 5.2) - (4 \cdot 6.2 \cdot 4)$$

$$|M_a| = 447.4 - 436$$

$$|M_a| = 11.4$$

$$M_a = \begin{bmatrix} 5.2 & 4 & 7 \\ 6.2 & 4 & 8 \\ 4.6 & 5 & 4 \end{bmatrix} \begin{bmatrix} 5.2 & 4 & 7 \\ 6.2 & 4 & 8 \\ 4.6 & 5 & 4 \end{bmatrix}$$

$$\underline{|M_a| = 11.4}$$

$$M_b = \begin{bmatrix} 4 & 5.2 & 7 \\ 6 & 6.2 & 8 \\ 5 & 4.6 & 4 \end{bmatrix}$$

$$M = \begin{bmatrix} 4 & 4 & 7 \\ 6 & 4 & 8 \\ 5 & 5 & 4 \end{bmatrix}$$

$$M_c = \begin{bmatrix} 4 & 4 & 5.2 \\ 6 & 4 & 6.2 \\ 5 & 5 & 4.6 \end{bmatrix}$$

$$|M| = 38$$

$$|M_b| = (4 \cdot 6.2 \cdot 4) + (5.2 \cdot 8 \cdot 5) + (7 \cdot 6 \cdot 4.6)$$

$$-(5 \cdot 6.2 \cdot 7) - (4.6 \cdot 8 \cdot 4) - (4 \cdot 6 \cdot 5.2)$$

$$|M_b| = 500.4 - 489$$

$$|M_b| = 11.4$$

$$M = \begin{bmatrix} 4 & 4 & 7 \\ 6 & 4 & 8 \\ 5 & 5 & 4 \end{bmatrix}$$

$$\underline{|M| = 38}$$

$$M_a = \begin{bmatrix} 5.2 & 4 & 7 \\ 6.2 & 4 & 8 \\ 4.6 & 5 & 4 \end{bmatrix}$$

$$\underline{|M_a| = 11.4}$$

$$M_b = \begin{bmatrix} 4 & 5.2 & 7 \\ 6 & 6.2 & 8 \\ 5 & 4.6 & 4 \end{bmatrix} \begin{bmatrix} 4 & 5.2 & 7 \\ 6 & 6.2 & 8 \\ 5 & 4.6 & 4 \end{bmatrix}$$

$$\underline{|M_b| = 11.4}$$

$$M_c = \begin{bmatrix} 4 & 4 & 5.2 \\ 6 & 4 & 6.2 \\ 5 & 5 & 4.6 \end{bmatrix}$$

$$|M_c| = (4 \cdot 4 \cdot 4.6) + (4 \cdot 6.2 \cdot 5) + (5.2 \cdot 6 \cdot 5)$$

$$-(5 \cdot 4 \cdot 5.2) - (5 \cdot 6.2 \cdot 4) - (4.6 \cdot 6 \cdot 4)$$

$$|M_c| = 353.6 - 338.4$$

$$|M_c| = 15.2$$

$$M = \begin{bmatrix} 4 & 4 & 7 \\ 6 & 4 & 8 \\ 5 & 5 & 4 \end{bmatrix}$$

$$\underline{|M| = 38}$$

$$M_a = \begin{bmatrix} 5.2 & 4 & 7 \\ 6.2 & 4 & 8 \\ 4.6 & 5 & 4 \end{bmatrix}$$

$$\underline{|M_a| = 11.4}$$

$$M_b = \begin{bmatrix} 4 & 5.2 & 7 \\ 6 & 6.2 & 8 \\ 5 & 4.6 & 4 \end{bmatrix}$$

$$\underline{|M_b| = 11.4}$$

$$M_c = \begin{bmatrix} 4 & 4 & 5.2 \\ 6 & 4 & 6.2 \\ 5 & 5 & 4.6 \end{bmatrix} \begin{bmatrix} 4 & 4 & 5.2 \\ 6 & 4 & 6.2 \\ 5 & 5 & 4.6 \end{bmatrix}$$

$$\underline{|M_c| = 15.2}$$

Cramer's Rule,

$$a = \frac{|M_a|}{|M|} \qquad b = \frac{|M_b|}{|M|} \qquad c = \frac{|M_c|}{|M|}$$

$$a = \frac{11.4}{38} \qquad b = \frac{11.4}{38} \qquad c = \frac{15.2}{38}$$

$$\boxed{a = 0.3} \qquad \boxed{b = 0.3} \qquad \boxed{c = 0.4}$$

ans *ans* *ans*

$$M = \begin{bmatrix} 4 & 4 & 7 \\ 6 & 4 & 8 \\ 5 & 5 & 4 \end{bmatrix}$$

$$\underline{|M| = 38}$$

$$M_a = \begin{bmatrix} 5.2 & 4 & 7 \\ 6.2 & 4 & 8 \\ 4.6 & 5 & 4 \end{bmatrix}$$

$$\underline{|M_a| = 11.4}$$

$$M_b = \begin{bmatrix} 4 & 5.2 & 7 \\ 6 & 6.2 & 8 \\ 5 & 4.6 & 4 \end{bmatrix}$$

$$\underline{|M_b| = 11.4}$$

$$M_c = \begin{bmatrix} 4 & 4 & 5.2 \\ 6 & 4 & 6.2 \\ 5 & 5 & 4.6 \end{bmatrix}$$

$$\underline{|M_c| = 15.2}$$

SEATWORK